

Honors Analysis

Final exam

December 11th, 2025

Question 1. (25pts) Write explicit examples of the following. One is not possible. No justification is required.

- (a) A closed and bounded set that is not compact in some metric space.
- (b) A C^1 bijection f between two sets A and B in $\mathbb{R}^n$ so that $Df(x)$ is invertible for all $x \in A$, but the inverse function f^{-1} is not C^1 .
- (c) A compact subset of $C([0, 1])$ (with the sup-norm) so that for any pair of distinct points x and y in $[0, 1]$, there is a function f in the set such that $f(x) \neq f(y)$.
- (d) An ODE of the form $x'(t) = f(x(t))$ with $x(0) = x_0$ and $f : \mathbb{R} \rightarrow \mathbb{R}$ a C^1 function, that has a solution in some interval $t \in (-a, b)$ but not on the full real line $t \in \mathbb{R}$.
- (e) A differential form ω such that $\omega \wedge \omega \wedge \omega \neq 0$.

Question 2. (25pts) Let M be a metric space so that every closed and bounded subset is compact. Prove that M is complete.

Hint. It was a homework problem.

Question 3. (25pts) Suppose that $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is differentiable and its derivative satisfies $\|Df(x)\| < 1$ for all $x \in \mathbb{R}^n$ (here $\|\cdot\|$ stands for the operator norm of the linear transformation $Df(x)$).

- (a) Is f a weak contraction?
- (b) Does it have a fixed-point?
- (c) What if we assume in addition that f is bounded. Would it have a fixed point then?

Hint. There was a similar homework problem in 1D. I made it higher dimensional and added part (c) to make it more interesting.

Question 4. (25pts) Let ω be a 2-form in $\mathbb{R}^3 \setminus \{0\}$ (i.e. ω is defined everywhere except at the origin). Assume that $d\omega = 0$ in $\mathbb{R}^3 \setminus \{0\}$.

- Prove that the integral of ω over any sphere centered at the origin is the same (that is, its integral does not depend on the radius of the sphere).
- Consider

$$\omega = \frac{1}{(x^2 + y^2 + z^2)^{3/2}}(x \, dy \wedge dz + y \, dz \wedge dx + z \, dx \wedge dy).$$

Verify that in this example $d\omega = 0$ in $\mathbb{R}^3 \setminus \{0\}$ and the integral of ω over the unit sphere S^2 is not zero.